Chapter 2 in [1]

Exercise (2.1-7)**.** Let a random experiment be the casting of a pair of unbiased six-sided dice, and let X equal the smaller of the outcomes if they are different and the common value if they are equal.

1. With reasonable assumptions, find the p.m.f. of X.
2. Let Y equal the range of the two outcomes (i.e. the absolute value of the difference of the largest and the smallest outcomes). Determine the p.m.f. $g(y)$ of Y for $y = 0, 1, 2, 3, 4, 5$.

Solution. 題目表示出 X 是六面骰骰兩次中最小點數的隨機變數。因爲合計有 *36* 種可能狀況，其中可以知道數字爲 *1* 的情況有 *11* 種，以此類推。更具體一點爲，考慮 $X = x$，那麼我們可以將

$$P(X \geq x) - P(X \geq x + 1) = P(X = x)$$

用來計算，因此

$$(7 - x)^2 - (6 - x)^2 = 13 - 2x$$

所以 X 的 *p.m.f.* 爲

$$\frac{13 - 2x}{36}$$

而 Y 的 p.m.f. 則可以直接計數得到

$$P(Y = y) = \begin{cases} \frac{1}{6}, & y = 0, \\ \frac{6-y}{18}, & y = 1, 2, 3, 4, 5. \end{cases}$$

1 References

[1] R. V. Hogg and E. A. Tanis, *Probability and Statistical Inference*, Pearson, 2010.